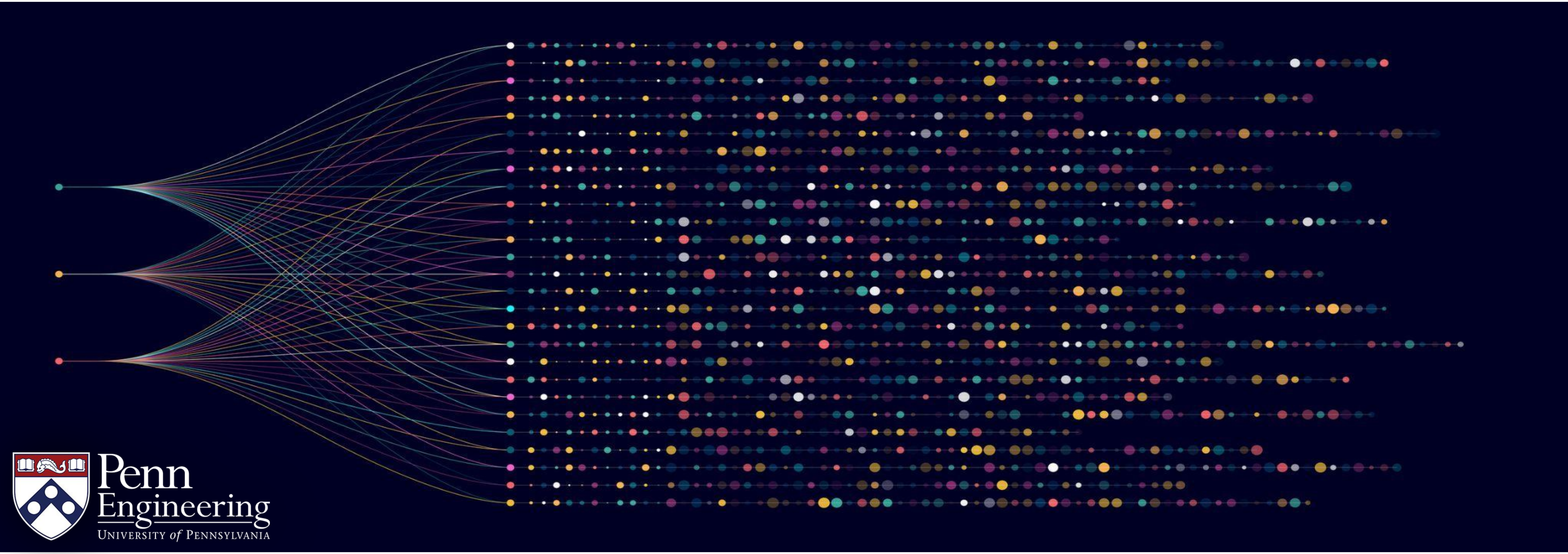
Algorithms for Big Data

Anindya De Erik Waingarten

Random variables



Big data and probability

- Randomized algorithms are fundamental to modern data analysis

- Both the design and analysis of such algorithms crucially rely on probabilistic analysis

- Good grasp of the basics of probability theory is crucial for success!

What are random variables?

Suppose we perform some probabilistic experiment – example toss three fair coins

A random variable is a function whose value is completely determined by the outcome of the experiment – e.g., in the three coin tosses, the number of heads is a random variable

Example of random variable

Example – Toss three fair coins (fair coin means $\Pr[H] = \Pr[T] = 1/2$)

Space of outcomes is {HHH}, {HHT}, {HTH}, {HTT}, {THH}, {THT}, {TTH} and {TTT}

(each with probability 1/8)

Let **Count** denote the random variable which counts the number of heads in three coin tosses

$$\mathbf{Count} = \begin{cases} 3 \text{ w.p. } 1/8 \\ 2 \text{ w.p. } 3/8 \\ 1 \text{ w.p. } 3/8 \\ 0 \text{ w.p. } 1/8 \end{cases}$$

Random variables -- continued

We will mostly think of discrete valued random variables – i.e., our random variables take values from a discrete set (in fact, mostly a finite set)

For a random variable $\boldsymbol{X}$**,** we use $\Pr[\boldsymbol{X}=x]$ to denote the probability that the random variable $\boldsymbol{X}$ takes the value x

Note that $\sum_x \Pr[\boldsymbol{X}=x] = 1$. In other words, the sum of the probabilities over the various outcomes x is 1.

Expectation of a random variable

A fundamental quantity associated with a random variable is its expectation

We define the expectation of a random variable $\boldsymbol{X}$ as $E[\boldsymbol{X}] := \sum_x x \cdot \Pr[\boldsymbol{X} = x]$.

Note that the expectation of random variable $\boldsymbol{X}$ is its "average value" over the space of outcomes

Note: The expectation of a random variable $\boldsymbol{X}$ is also sometimes referred to as its mean

Expectation of a random variable – an example

Let us see an example of expectation calculation – in particular, consider **Count**, the random variable for counting the # of heads in three fair coin tosses.

$$\mathbf{Count} = \begin{cases} 3 \text{ w.p. } 1/8 \\ 2 \text{ w.p. } 3/8 \\ 1 \text{ w.p. } 3/8 \\ 0 \text{ w.p. } 1/8 \end{cases}$$

$$E[\mathbf{Count}] = 3 \cdot {}^{1}\!/_{8} + 2 \cdot {}^{3}\!/_{8} + 1 \cdot {}^{3}\!/_{8} + 0 \cdot {}^{1}\!/_{8} = {}^{3}\!/_{2} \ .$$

Linearity of expectation

Arguably, the most fundamental fact about expectations of random variables is linearity of expectation.

Namely, for any two random variables $\boldsymbol{X}$ and $\boldsymbol{Y}$,

I. $E[\boldsymbol{X}+\boldsymbol{Y}] = E[\boldsymbol{X}] + E[\boldsymbol{Y}]$ and
II. For any $c \in \mathbb{R}$, $E[c \cdot \boldsymbol{X}] = c \cdot E[\boldsymbol{X}]$.

Linearity of expectation – a powerful tool

Linearity of expectation is a very powerful tool to compute expectations of random variables – even very complicated ones.

Example: Let $A = \{1,2,3,\ldots,n\}$. Let $\boldsymbol{B}$ be a uniformly random permutation of A **–** i.e., a uniformly random rearrangement of A

So, if $n = 3$, then with probability $1/6$ each, $\boldsymbol{B} = \{1,2,3\}; \{1,3,2\}; \{2,1,3\}; \{2,3,1\}; \{3,1,2\}; \{3,2,1\}$

Fixed points of a permutation

For any permutation, the fixed points in a permutation are positions which are ”unchanged”

Example: Permutation $\{1,2,3\}$ -- fixed points are 1, 2, and 3

Permutation $\{1,3,2\}$ -- fixed point is 1

Permutation $\{2,1,3\}$ -- fixed point is 3

Permutation $\{2,3,1\}$ -- no fixed points

Permutation $\{3,2,1\}$ -- fixed point is 2

Permutation $\{3,1,2\}$ -- no fixed points

Number of fixed points of a random permutation

Let $A = \{1,2,3,\ldots,n\}$. Let $\boldsymbol{B}$ be a uniformly random permutation of A. Let $\mathrm{FP}(\boldsymbol{B})$ denote the number of fixed points in $\boldsymbol{B}$.

Compute $E[\mathrm{FP}(\boldsymbol{B})]$ – i.e., the expected number of fixed points of random permutation

Number of fixed points of a random permutation: attempt #1

Let $A = \{1,2,3,\ldots,n\}$. Let $\boldsymbol{B}$ be a uniformly random permutation of A. Let $\mathrm{FP}(\boldsymbol{B})$ denote the number of fixed points in $\boldsymbol{B}$.

Note that by the definition of expectation:

$E[\mathrm{FP}(\boldsymbol{B})] = \sum_{i=0}^{n} i \cdot \Pr[\mathrm{FP}(\boldsymbol{B}) = i]$.

Complication: How to compute $\Pr[\mathrm{FP}(\boldsymbol{B}) = i]$?

Attempt #2: exploiting linearity of expectation

Let $A = \{1,2\,,3,\ldots,n\}$. Let $\boldsymbol{B}$ be a uniformly random permutation of A. Let $\mathrm{FP}(\boldsymbol{B})$ denote the number of fixed points in $\boldsymbol{B}$.

Let $\mathrm{FP}_j(\boldsymbol{B})$ denote the random variable which is 1 if j is a fixed point of $\boldsymbol{B}$ and 0 otherwise

Key observation: $\mathrm{FP}(\boldsymbol{B}) = \sum_{j=1}^{\boldsymbol{n}} FP_j(\boldsymbol{B})$

Linearity of expectation implies $E[\mathrm{FP}(\boldsymbol{B})] = \sum_{j=1}^{\boldsymbol{n}} E[FP_j(\boldsymbol{B})]$ $\quad - \quad$ (1)

Computing $E[FP_j(\boldsymbol{B})]$

Let $\mathrm{FP}_j(\boldsymbol{B})$ denote the random variable which is 1 if j is a fixed point of $\boldsymbol{B}$ and 0 otherwise

$$E[FP_j(\boldsymbol{B})] = 1 \cdot \Pr[\mathrm{FP}_j(\boldsymbol{B}) = 1] + 0 \cdot \Pr[\mathrm{FP}_j(\boldsymbol{B}) = 0] = \Pr[\mathrm{FP}_j(\boldsymbol{B}) = 1]$$

Sidenote: A random variable $\boldsymbol{X}$ which takes only two values 0 and 1 is called an indicator (or Bernoulli) random variable

For such an $\boldsymbol{X}$, $E[\boldsymbol{X}] = 1 \cdot \Pr[\boldsymbol{X} = 1] + 0 \cdot \Pr[\boldsymbol{X} = 0] = \Pr[\boldsymbol{X} = 1]$

Computing $E[FP_j(\boldsymbol{B})]$

Let $\mathrm{FP}_j(\boldsymbol{B})$ denote the random variable which is 1 if j is a fixed point of $\boldsymbol{B}$ and 0 otherwise

$E[FP_j(\boldsymbol{B})] = \Pr[\mathrm{FP}_j(\boldsymbol{B}) = 1]$

For a random permutation $\boldsymbol{B}$, the index j is equally likely to get mapped to any of the positions in $\{1,2, \dots, n\}$. Thus, $\Pr\left[\mathrm{FP}_j(\boldsymbol{B}) = 1\right] = \frac{1}{n}$

This implies for any j, $E[FP_j(\boldsymbol{B})] = \frac{1}{n}$

Computing $E[\mathrm{FP}(\boldsymbol{B})]$

Recall equation (1) -- $E[\mathrm{FP}(\boldsymbol{B})] = \sum_{j=1}^{n} E[FP_j(\boldsymbol{B})]$

As each $E[FP_j(\boldsymbol{B})] = \frac{1}{n}$ for each index j, it follows that $E[\mathrm{FP}(\boldsymbol{B})] = 1$

Thus, for a random permutation $\boldsymbol{B}$ of the set $A = \{1,2,3,\ldots,n\}$, the expected number of fixed points is 1 (does not depend on n)

Fun way to reinterpret our calculation

Suppose n guests come to a party on a rainy afternoon. Each come with their umbrella

They put their umbrellas in an umbrella bucket. At the end of the party, the guests just randomly choose an umbrella from the bucket to take back with them ☺

Viewing the mapping between the umbrella and the guests as a random permutation of $A = \{1,2,3,\ldots,n\}$, it means "on an average", exactly 1 guest will get back their own umbrella.

Summary

- Random variables are functions of the underlying probabilistic experiment

- Also, we're looking at real-valued random variables (i.e., functions take value in $\mathbb{R}$) though generalizations are possible

- For a random variable $\boldsymbol{X}$, its expectation is the average value it takes over all possible outcomes of the underlying experiment

Summary

- Linearity of expectation:

$$E[\boldsymbol{X} + \boldsymbol{Y}] = E[\boldsymbol{X}] + E[\boldsymbol{Y}]$$

- Useful tool: Split a random variable $\boldsymbol{X}$ as

  $\boldsymbol{X} = \boldsymbol{Z}_1 + \cdots + \boldsymbol{Z}_n$ where each $\boldsymbol{Z}_i$ is "simple";

  Analyze $E[\boldsymbol{Z}_i]$ individually and obtain

$$E[\boldsymbol{X}] = E[\boldsymbol{Z}_1] + \cdots + E[\boldsymbol{Z}_n]$$

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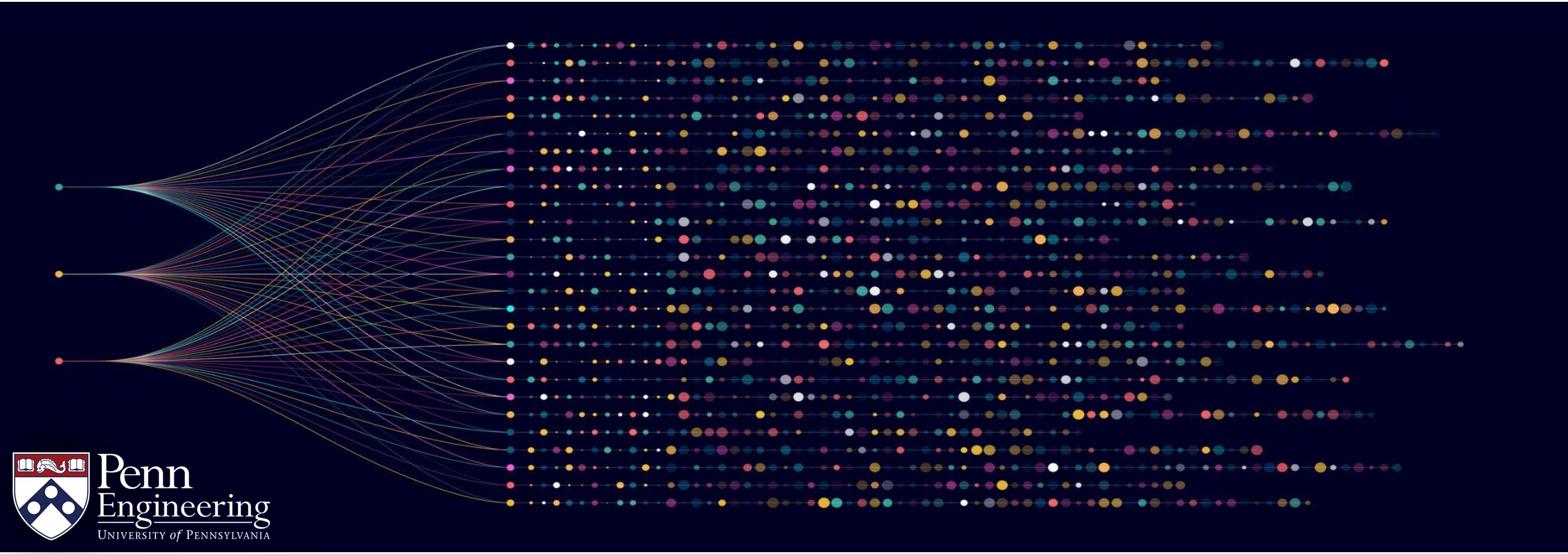
Independence of random variables



An old saying

- Old saying: “Lightning never strikes the same place twice”
- Clearly False: There are lightning prone areas

Independent events?

- Lightning affected by temperature, moisture, altitude etc. -- so areas prone to lightning often remain so for extended durations

Independence of factors?

Implicitly, this saying assumes that lightning strikes are **independent events** – an assumption which is clearly false!

Arguably, independence of events is the most (mis)used assumption – both in theory and practice

Examples: 2008 financial crisis – mathematical models largely ignored correlations between investors

What is independence?

Formally, two random variables $\boldsymbol{A}$ and $\boldsymbol{B}$ are said to be independent if for every possible choice of x and y, we have that:

$$\Pr[\boldsymbol{A} = x \textit{ and } \boldsymbol{B} = y] = \Pr[\boldsymbol{A} = x] \cdot \Pr[\boldsymbol{B} = y].$$

More generally, a set of random variables $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_n$ are mutually independent if for every choice of $x_1, \ldots, x_n$, it holds that

$$\Pr[\boldsymbol{A_1} = x_1, \boldsymbol{A_2} = x_2, \ldots, \boldsymbol{A_n} = x_n] = \prod_{i=1}^{n} \Pr[\boldsymbol{A_i} = x_i].$$

Why independence?

Succinct representation: If the variables $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_n$ are mutually independent, then the joint distribution of $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_n$ can be described using just the individual random variables $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_n$.

Example: Suppose $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_n$ are Bernoulli random variables, i.e., take values in $\{0,1\}$.

Useful notation: $\{0,1\}^n$ denotes the set of n –tuples whose coordinates comes from $\{0,1\}$.

(For example, for n = 2, $\{0,1\}^2 = \{(0,0), (0,1), (1,0), (1,1)\}$)

Note that $(\boldsymbol{A}_1, \ldots, \boldsymbol{A}_n)$ takes values in $\{0,1\}^n$.

Exponential description of joint probability distributions

If $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_n$ are not independent, then in the worst case, specifying the joint distribution of $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_n$ would require specifying $\Pr[\boldsymbol{A}_1 = x_1, \boldsymbol{A}_2 = x_2, \ldots, \boldsymbol{A}_n = x_n]$ at every point $(x_1, \ldots, x_n)$

In the worst case, that will take specifying $\Theta(2^n)$ real numbers

∗ For those who are not familiar with the big Theta notation, f(n)= Θ(g(n)) is equivalent to f(n) = O(g(n)) and f(n) = Ω(g(n)).

Succinct size for independent random variables

Suppose $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_n$ are independent Bernoulli random variables such that for each $1 \leq i \leq n$, $\Pr[\boldsymbol{A}_i = 1] = \rho_i$ and $\Pr[\boldsymbol{A}_i = 0] = 1 - \rho_i$. Succinctly, $\Pr[\boldsymbol{A}_i = x_{\boldsymbol{i}}] = 1 - \rho_i + x_{\boldsymbol{i}} \cdot (2\rho_i - 1)$.

$(\boldsymbol{A}_1, \ldots, \boldsymbol{A}_n)$ can take values $(x_1, \ldots, x_n) \in \{0,1\}^n$ and

$$\Pr[\boldsymbol{A}_1 = x_1, \boldsymbol{A}_2 = x_2, \ldots, \boldsymbol{A}_n = x_n] = \prod_{i=1}^{n} \Pr[\boldsymbol{A}_i = x_i] = \prod_{i=1}^{n}(1 - \rho_i + x_{\boldsymbol{i}} \cdot (2\rho_i - 1)).$$

Summary: If $\boldsymbol{A}_1, \ldots, \boldsymbol{A}_n$ are independent, then the joint distribution of $(\boldsymbol{A}_1, \ldots, \boldsymbol{A}_n)$ can be specified using n real numbers $\rho_1, \ldots, \rho_n$.

Mathematically tractable behavior

Mathematically tractable behavior: Independent random variables are easy to analyze – many of the most fundamental results in probability such as the Central limit theorem or Chernoff bounds require independence

Theorems in probability theory which require no independence exist but are quite rare

What makes random variables independent?

Physical independence of underlying actions: If the physical action governing the random variables are independent, then the random variables are independent

Example: If we toss 2 coins *independently,* then the outcome of the first coin (H, T) is is independent of the outcome of the second coin (H, T)

What makes random variables independent?

Subtle examples of independence: Sometimes random variables can be independent even when the underlying physical actions are quite dependent

Example: roll two fair six-sided dice independently

$\boldsymbol{X} = \mathbf{1}$ if the first die is (2, 4, 6) and $\boldsymbol{X} = \mathbf{0}$ otherwise

$\boldsymbol{Y} = \mathbf{1}$ if the sum of the dice is 7 and $\boldsymbol{Y} = \mathbf{0}$ otherwise

Seemingly dependent (because the sum of the dice is dependent on the outcome of the first) but turns out that $\boldsymbol{X}$ and $\boldsymbol{Y}$ are independent Verify: left as an exercise

Summary

- There are many other ”subtle” examples of independence – e.g., in the context of continuous random variables, stochastic processes

- See some examples later in the course

- Independence of random variables important in both theory and practice

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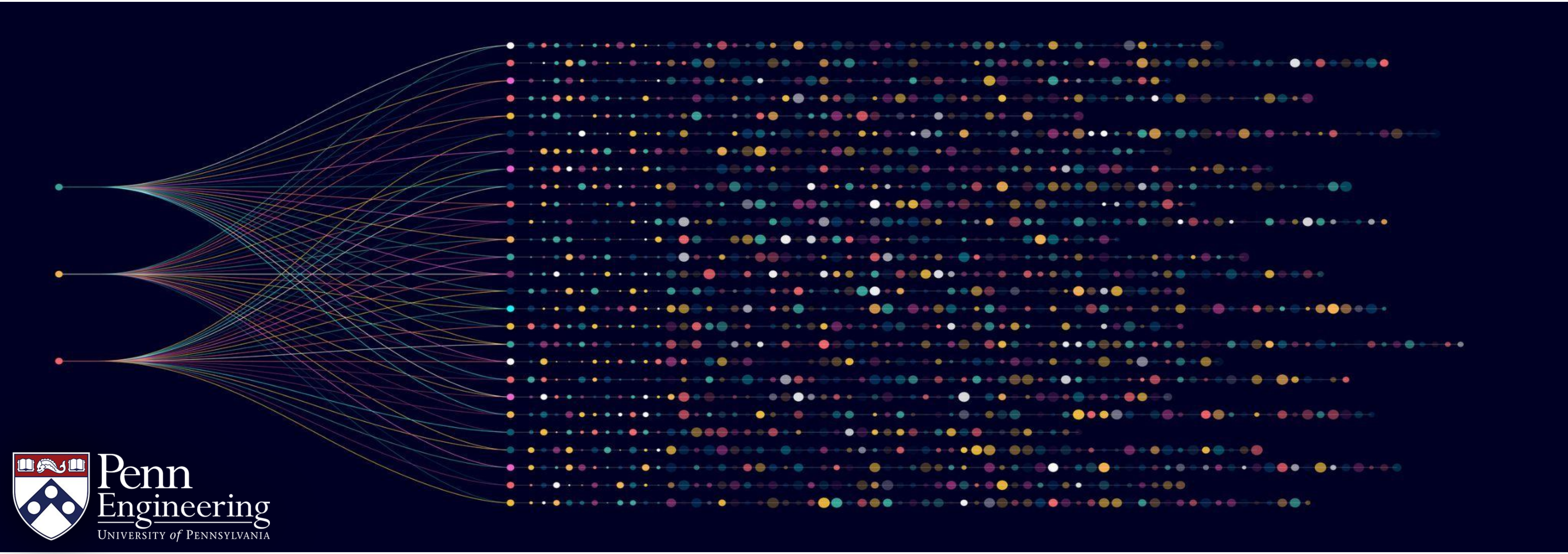
Markov's inequality



How big is the median household income?

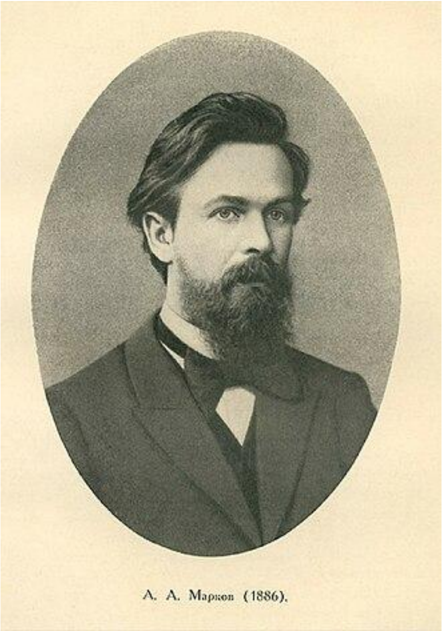
- US Census Bureau: The per capita income in the US is ~ $38,000

$$\frac{\text{Total annual income by everyone in the US}}{\text{US population}} \approx \$38{,}000.$$

Question: How big can the median income be?

Is it possible that 50% or more people make $80,000 or more?

Markov’s inequality



Markov’s inequality states that for any random variable $\boldsymbol{X}$ which is always non-negative and $t > 0$, $\Pr[\boldsymbol{X} \geq t] \leq \frac{E[\boldsymbol{X}]}{t}$

Corollary: if $\boldsymbol{X}$ is non-negative, then $\Pr[\boldsymbol{X} \geq 10E[\boldsymbol{X}]] \leq {}^{1}/_{10}$

Markov’s inequality is fundamental in probability theory – does not require any assumptions beyond non-negativity of $\boldsymbol{X}$

Proof of Markov's inequality

Markov's inequality: If $\boldsymbol{X}$ which is always non-negative and $t > 0$, $\Pr[\boldsymbol{X} \geq t] \leq \frac{E[\boldsymbol{X}]}{t}$.

Proof: By definition, $E[\boldsymbol{X}] = \sum_x x \cdot \Pr[\boldsymbol{X} = x]$.

$E[\boldsymbol{X}] = \sum_{x<t} x \cdot \Pr[\boldsymbol{X} = x] + \sum_{x \geq t} x \cdot \Pr[\boldsymbol{X} = x] \quad -(1)$

Observe that $\sum_{x<t} x \cdot \Pr[\boldsymbol{X} = x] \geq 0$ – since $\boldsymbol{X}$ can only take non-negative values and $\Pr[\boldsymbol{X} = x]$ is always non-negative

Thus, $E[\boldsymbol{X}] \geq \sum_{x \geq t} x \cdot \Pr[\boldsymbol{X} = x]$.

Proof of Markov's inequality

Markov's inequality: If $\boldsymbol{X}$ which is always non-negative and $t > 0$, $\Pr[\boldsymbol{X} \geq t] \leq \frac{E[\boldsymbol{X}]}{t}$.

Proof (continued):

So, fat we have $E[\boldsymbol{X}] \geq \sum_{x \geq t} x \cdot \Pr[\boldsymbol{X} = x]$ $-$ (2)

$\sum_{x \geq t} x \cdot \Pr[\boldsymbol{X} = x] \geq \sum_{x \geq t} t \cdot \Pr[\boldsymbol{X} = x] = t \cdot (\sum_{x \geq t} \Pr[\boldsymbol{X} = x])$ $-$ (3).

However, $(\sum_{x \geq t} \Pr[\boldsymbol{X} = x]) = \Pr[\boldsymbol{X} \geq t]$. Plugging into (3), $\sum_{x \geq t} x \cdot \Pr[\boldsymbol{X} = x] \geq t \cdot \Pr[\boldsymbol{X} \geq t]$.

Plugging back into (2), we get $E[\boldsymbol{X}] \geq t \cdot \Pr[\boldsymbol{X} \geq t]$ (same as our claim)

Application of Markov's inequality

Application #1: Relating Average and Median income

Let the population of a country be N and the incomes be $x_1, \ldots, x_N \geq 0$.

Let $\boldsymbol{Z}$ be the random variable defined as: for each $1 \leq i \leq N$, with probability $\frac{1}{N}$, $\boldsymbol{Z} = x_i$.

Note that $\boldsymbol{Z} \geq 0$ and $\mathrm{E}[\boldsymbol{Z}] = P$ – per capita income

By Markov's inequality, $\Pr[\boldsymbol{Z} \geq 2P] \leq \frac{\mathrm{E}[\boldsymbol{Z}]}{2P} = \frac{1}{2}$.

Note that if the median income of the country is M^*, then $\Pr[\boldsymbol{Z} \geq M^*] = \frac{1}{2}$.
Thus, the median income $M^* \leq 2P$.

So, if the US per capita income is $38,000, the median income cannot exceed $76,000

Application #2: Number of fixed points of a random permutation

Recall the setting: $A = \{1,2, \ldots, n\}$ and let $\boldsymbol{B}$ be a random permutation of A.

We are interested in the number of fixed points of $\boldsymbol{B}$ (denoted by $\mathrm{FP}(\boldsymbol{B})$)

– i.e., positions i which are unchanged in $\boldsymbol{B}$

Previous video: We established $\mathrm{E}[\mathrm{FP}(\boldsymbol{B})] = 1$

We now notice that $\mathrm{FP}(\boldsymbol{B})$ is a non-negative random variable

Thus, Markov's inequality implies that for $c \geq 1$, $\Pr[\mathrm{FP}(\boldsymbol{B}) \geq c] \leq \frac{\mathrm{E}[\mathrm{FP}(\boldsymbol{B})]}{c} = \frac{1}{c}$.

Aside: Montmort [1700s] established that $\Pr[\mathrm{FP}(\boldsymbol{B}) = c] \sim \frac{1}{e \cdot c!}$

Summary

Markov's inequality – extremely powerful tool in probability theory

Can be applied in the most general of situations – only requirement is that the random variable is non-negative

Other strong probability bounds often obtained by applying Markov's inequality on the "right random variable"

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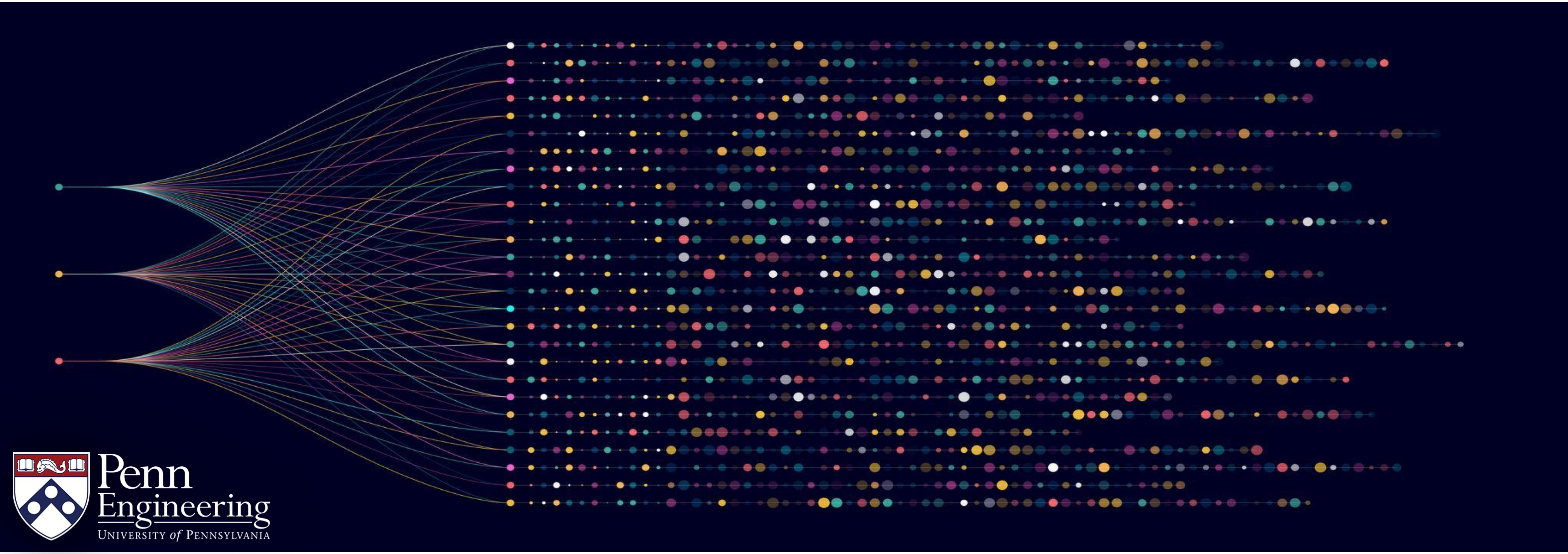
Variance and Chebyshev's inequality



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An anecdote about averages

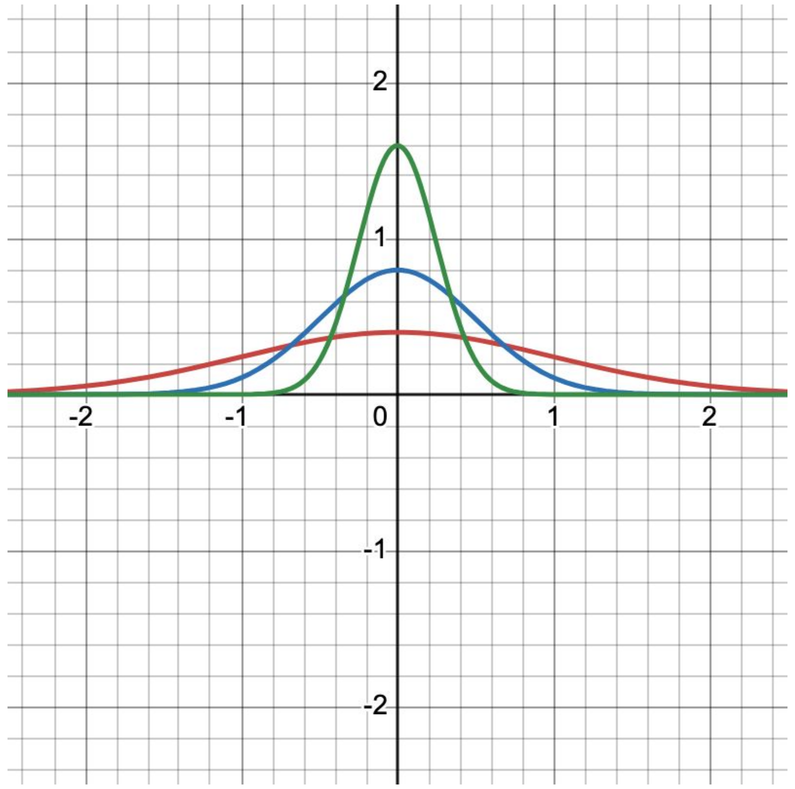
Famous computer scientist and Columbia CS Professor Christos Papadimitriou has ~ 300 co-authors. Average worth of his co-authors is nearly 400 million dollars.

An anecdote about averages

Christos wrote a paper in the mid 70s with a then undergraduate student – Bill Gates (today worth ~ $120 billion)

Averages can be misleading!

Moral of the story: averages can be misleading and do not tell the entire story for a distribution



Red curve has maximum spread and the green curve has minimum spread

Variance of a random variable

Given a real valued random variable $\boldsymbol{X}$, its variance is defined as

$$Var(\boldsymbol{X}) \coloneqq \mathrm{E}[(\boldsymbol{X} - \mu)^2] \text{ where } \mu \coloneqq \mathrm{E}[\boldsymbol{X}].$$

The variance measures the "average squared distance" of a random variable from its expectation

Note that $Var(\boldsymbol{X}) \geq 0$. In fact, the only way $Var(\boldsymbol{X}) = 0$ if $\boldsymbol{X}$ only takes one value.

Alternate way of writing variance

$Var(\boldsymbol{X}) \coloneqq \mathrm{E}[(\boldsymbol{X}-\mu)^2]$ where $\mu \coloneqq \mathrm{E}[\boldsymbol{X}]$.

$$\begin{aligned}\mathrm{E}[(\boldsymbol{X}-\mu)^2] &= \mathrm{E}[\boldsymbol{X}^2+\mu^2-2\mu\boldsymbol{X}] \\ &= \mathrm{E}[\boldsymbol{X}^2]+\mu^2-2\mathrm{E}[\mu\boldsymbol{X}] \text{ (by linearity of expectation)} \quad \text{- (1)}\end{aligned}$$

Now, $\mathrm{E}[\mu\boldsymbol{X}] = \mu\mathrm{E}[\boldsymbol{X}] = \mu^2$. Plugging this back into (1), we get
$\mathrm{E}[(\boldsymbol{X}-\mu)^2] = \mathrm{E}[\boldsymbol{X}^2]+\mu^2-2\mu^2 = \mathrm{E}[\boldsymbol{X}^2]-\mu^2$.

Thus, $Var(\boldsymbol{X}) = \mathrm{E}[\boldsymbol{X}^2]-\mu^2$ -- i.e.,
"gap between the expectation of the square and the square of the expectation"

Corollary: Since $Var(\boldsymbol{X}) \geq 0$, it follows that $\mathrm{E}[\boldsymbol{X}^2] \geq (\mathrm{E}[\boldsymbol{X}])^2$

Small variance ≈ Concentration of data

As $Var(\boldsymbol{X}) \coloneqq \mathrm{E}[(\boldsymbol{X} - \mu)^2]$, at a high level, $Var(\boldsymbol{X})$ measures how far $\boldsymbol{X}$ is from its mean

So, at an intuitive level, the smaller $Var(\boldsymbol{X})$ is, the more $\boldsymbol{X}$ is concentrated around its mean

Can we make this formal?

Chebyshev's inequality

Chebyshev's inequality: For any random variable $\boldsymbol{X}$,

$$\mathbf{Pr}[|\boldsymbol{X} - \mathrm{E}[\boldsymbol{X}]| \geq \mathrm{t}\,\sigma] \leq \frac{1}{\mathrm{t}^2},$$

where $\sigma^2 = Var(\boldsymbol{X})$. The quantity $\sigma = \sqrt{Var(\boldsymbol{X})}$ is referred to as the standard deviation of $\boldsymbol{X}$.

Intuitively: The standard deviation of $\boldsymbol{X}$, i.e., $\sqrt{Var(\boldsymbol{X})}$, determines the "scale" at which $\boldsymbol{X}$ typically deviates from its mean

As an example: If we plug in $\mathrm{t} = 5$, the probability that $\boldsymbol{X}$ deviates from $\mathrm{E}[\boldsymbol{X}]$ by more than 5σ is at most $\frac{1}{\mathrm{t}^2} = \frac{1}{5^2} = 4\%$

Proof of Chebyshev's inequality

Consider the random variable $\boldsymbol{Z} = (\boldsymbol{X} - \mathrm{E}[\boldsymbol{X}])^2$.

By definition, $\mathrm{E}[\boldsymbol{Z}] = \mathrm{Var}(\boldsymbol{X}) := \sigma^2$.

Since $\boldsymbol{Z}$ is a non-negative random variable, by Markov's inequality, for any $t \geq 0$,

$$\Pr\left[\boldsymbol{Z} \geq t^2\sigma^2\right] \leq \frac{\mathrm{E}[\boldsymbol{Z}]}{t^2\sigma^2} = \frac{\sigma^2}{t^2\sigma^2} = \frac{1}{t^2} \qquad - (1)$$

Observe that $\boldsymbol{Z} \geq t^2\sigma^2$ if and only if $|\boldsymbol{X} - \mathrm{E}[\boldsymbol{X}]| \geq t\sigma$.

Thus, $\Pr[|\boldsymbol{X} - \mathrm{E}[\boldsymbol{X}]| \geq t\sigma] = \Pr\left[\boldsymbol{Z} \geq t^2\sigma^2\right] \leq \frac{1}{t^2}$ (using (1)).

Proof of Chebyshev's inequality: in retrospect

Chebyshev's inequality is just Markov's inequality on the random variable $\boldsymbol{Z} = (\boldsymbol{X} - \mathrm{E}[\boldsymbol{X}])^2$

Many probabilistic bounds are obtained by Markov's inequality on the "right random variable". More examples later.

Summary

Variance is the gap between the ”expectation of the square” and “the square of the expectation”

Variance (and standard deviation) are measures of the spread of a random variable

Chebyshev’s inequality implies (for example) that a random variable deviates more than 5σ from its mean is at most 4%

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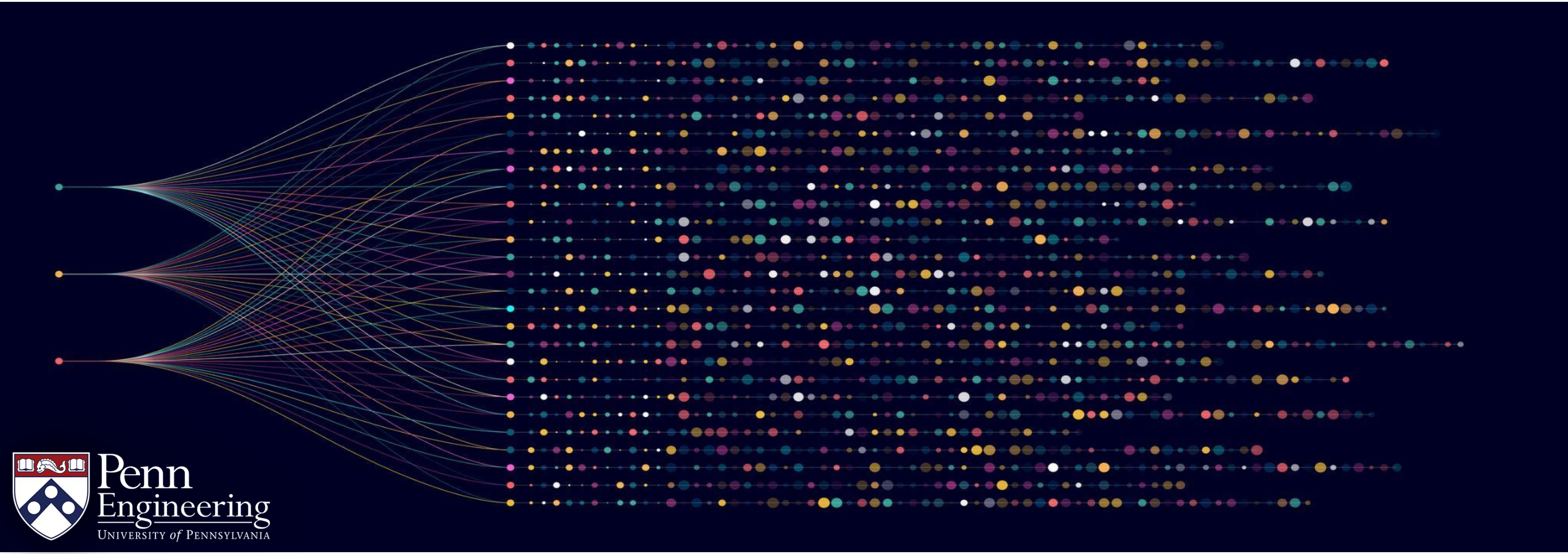
Continuous random variables



Continuous random variables

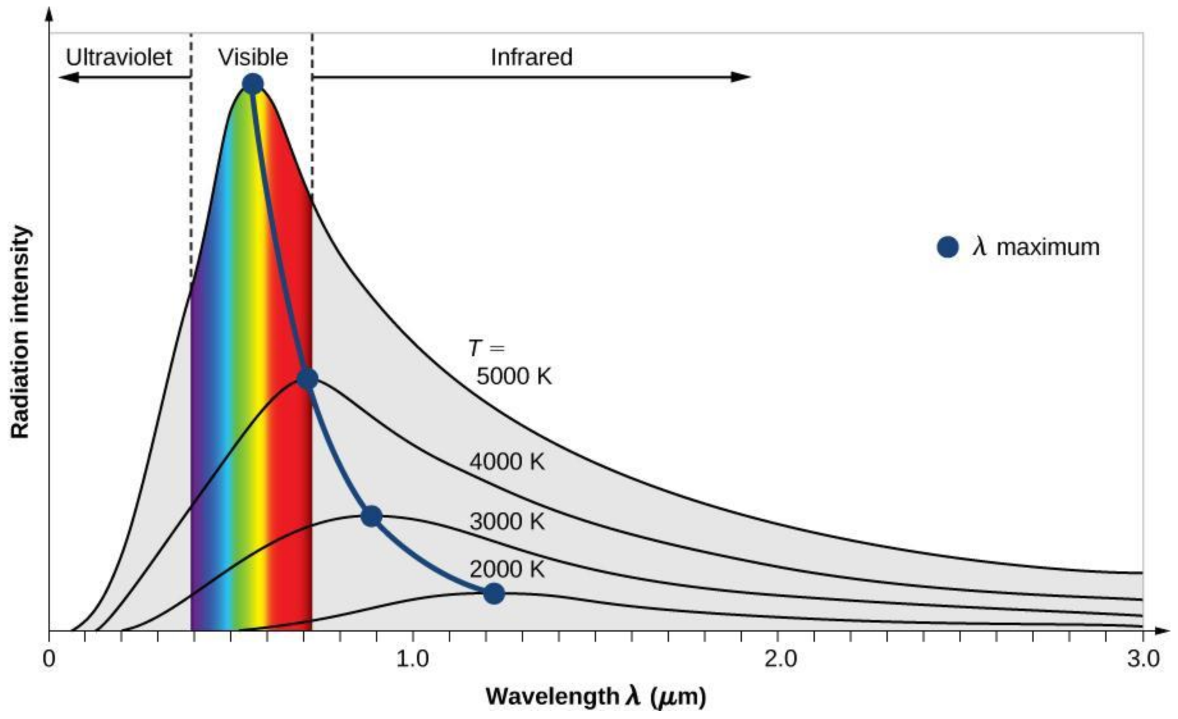
So far, our random variables have been discrete

Most of the course: suffices to think about discrete random variables

However, some parts of this course: necessary to study continuous random variables

Continuous random variables: where do they arise?

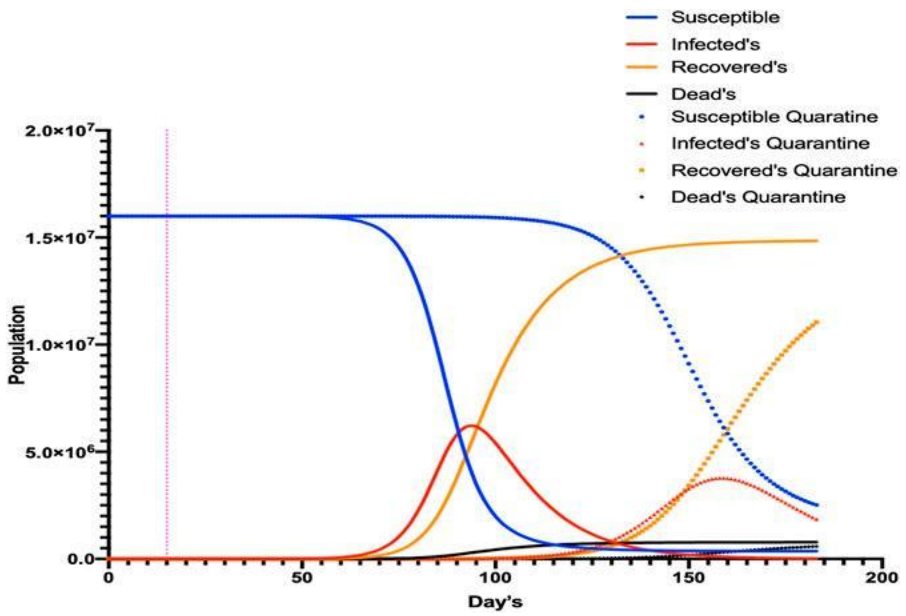
In theory and practice, many random variables are modeled as continuous random variables



Example #1: Planck's formula for blackbody radiation

Continuous random variables: where do they arise?

In theory and practice, many random variables are modeled as continuous random variables



Example #2: Models for spread of infectious diseases in epidemiology

Probability density function

Discrete random variables $\boldsymbol{X}$ are given by $\Pr[\boldsymbol{X} = z]$ for all points z in the support of $\boldsymbol{X}$

(support of $\boldsymbol{X}$ is the set of points z such that $\Pr[\boldsymbol{X} = z] \neq 0$)

For continuous random variables $\boldsymbol{X}$, meaningless to specify $\Pr[\boldsymbol{X} = z]$ **--** if $\boldsymbol{X}$ is continuous, the probability that it takes any specific value x is zero

Instead, we specify a continuous random variable using a probability density function $p(\cdot)$ (abbreviated as pdf)

Probability density function

For a continuous random variable $\boldsymbol{X}$, the pdf $p(\cdot)$ tells us the "density" of $\boldsymbol{X}$ at the point

For small δz, $\Pr\left[\boldsymbol{X} \in [z, z+\delta z]\right] \sim p(z) \cdot \delta z$.

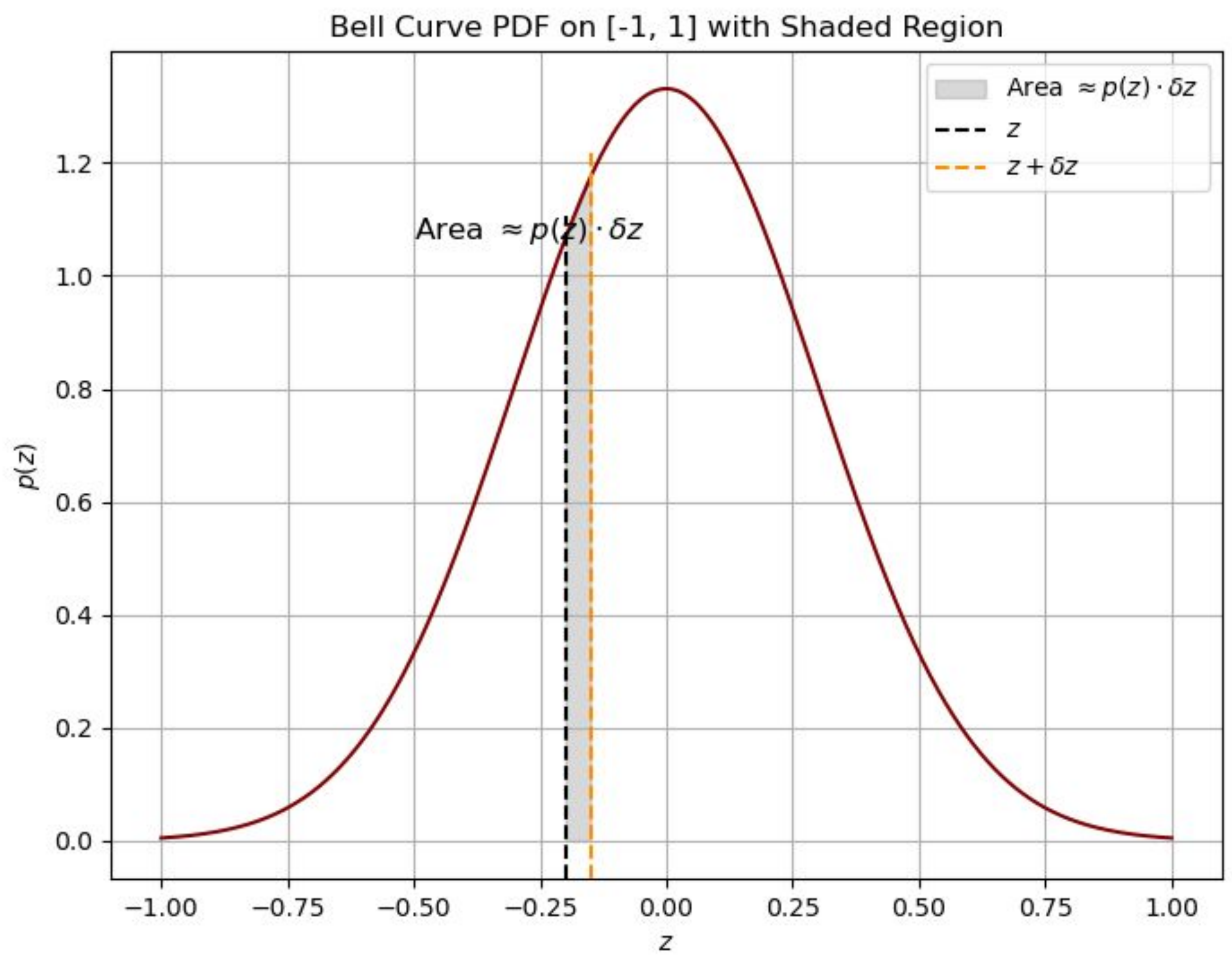
Or in the calculus sense,

$$\lim_{\delta z \to 0} \Pr\left[\boldsymbol{X} \in [z, z+\delta z]\right] = p(z) \cdot \delta z$$

If $p(\cdot)$ is given as a graph, then

$\Pr\left[\boldsymbol{X} \in [z, z+\delta z]\right] =$ area of gray shaded region

As $\delta z \to 0$, area of gray region = $p(z) \cdot \delta z$.



Probability density function

So far, we have considered $\Pr\big[\boldsymbol{X} \in [z, z + \delta z]\big]$, i.e., width of the interval $\delta z \to 0$

What happens in the general case? How do we get $\Pr\big[\boldsymbol{X} \in [z_1, z_2]\big]$ for general z_1, z_2?

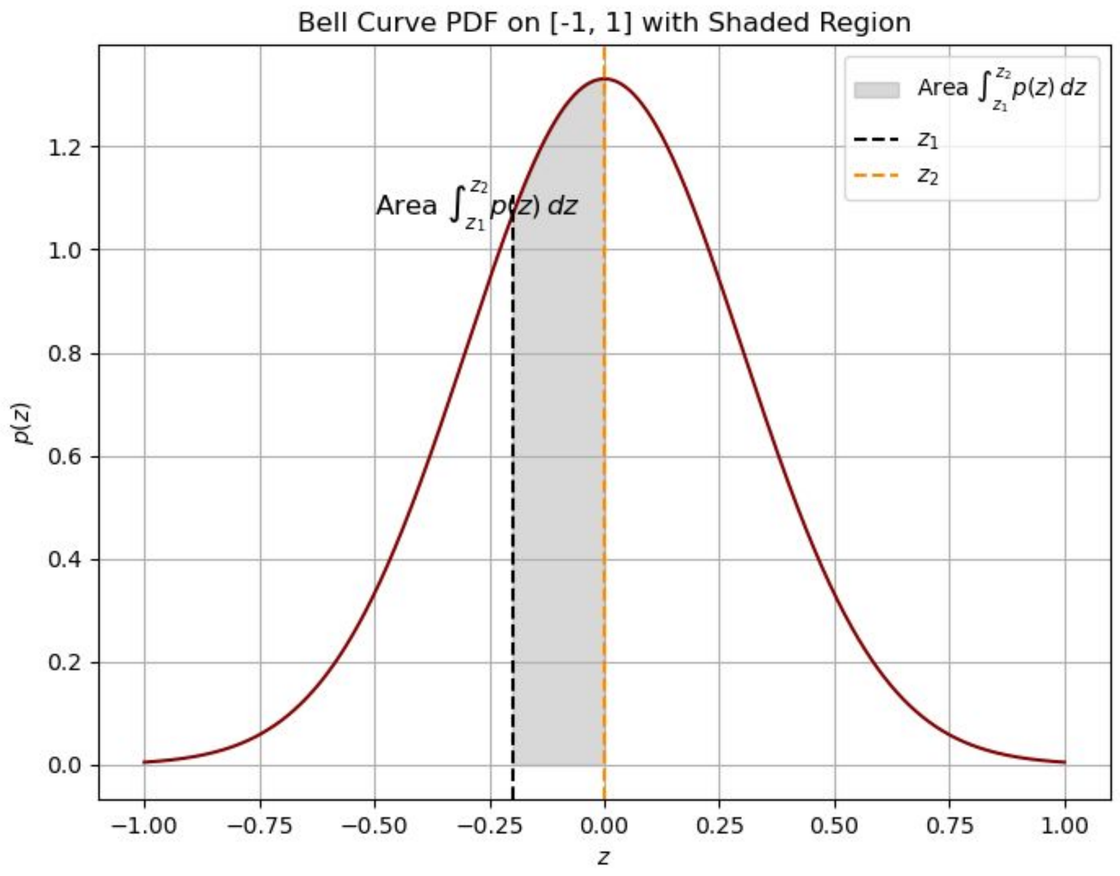
Obvious idea: break $[z_1, z_2]$ into many small intervals and add up the probabilities

In the language of calculus, $\Pr\big[\boldsymbol{X} \in [z_1, z_2]\big] = \int_{z_1}^{z_2} p(z) dz. \quad -(1)$

Probability density function

Thus, if the pdf of $\boldsymbol{X}$ is $p(\cdot)$,

$$\Pr\left[\boldsymbol{X} \in [z_1, z_2]\right] = \int_{z_1}^{z_2} p(z)dz. \quad -(1)$$



Dictionary between "discrete" and "continuous" worlds

For a continuous random variable $\boldsymbol{X}$

$$\Pr\left[\boldsymbol{X} \in [z_1, z_2]\right] = \int_{z_1}^{z_2} p(z)dz.$$

Compare it with the discrete world – For a random variable $\boldsymbol{X}$,

$$\Pr\left[\boldsymbol{X} \in [z_1, z_2]\right] = \sum_{z_1 \leq z \leq z_2} \Pr[\boldsymbol{X} = z].$$

Dictionary: 1. "Sum" gets replaced by an "integral"

2. $\Pr[\boldsymbol{X} = z]$ gets replaced by $p(z)$

Dictionary between "discrete" and "continuous" worlds

Discrete r. v.

Law of total probability:

$\sum_z \Pr[\boldsymbol{X} = z] = 1$

Expectation: $E[\boldsymbol{X}] \coloneqq \sum_z z \cdot \Pr[\boldsymbol{X} = z]$

Second moment: $E[\boldsymbol{X}^2] = \sum_z z^2 \cdot \Pr[\boldsymbol{X} = z]$

Continuous r. v.

Law of total probability:

$\int p(z)\, dz = 1$

Integral should go from $z = -\infty$ to ∞.

Can exclude sets where $p(z) = 0$.

Expectation: $E[X] \coloneqq \int z \cdot p(z)\, dz$

Second moment: $E[\boldsymbol{X}^2] = \int z^2 \cdot p(z)\, dz$

Dictionary between "discrete" and "continuous" worlds

Discrete r. v.

More generally,

$$E[f(\boldsymbol{X})] = \sum_z f(z) \cdot \Pr[\boldsymbol{X} = z].$$

Continuous r. v.

More generally,

$$E[f(\boldsymbol{X})] = \int f(z) \cdot p(z)\, dz.$$

Cumulative distribution function

Cumulative distribution function (cdf) for a continuous random variable $\boldsymbol{X}$:

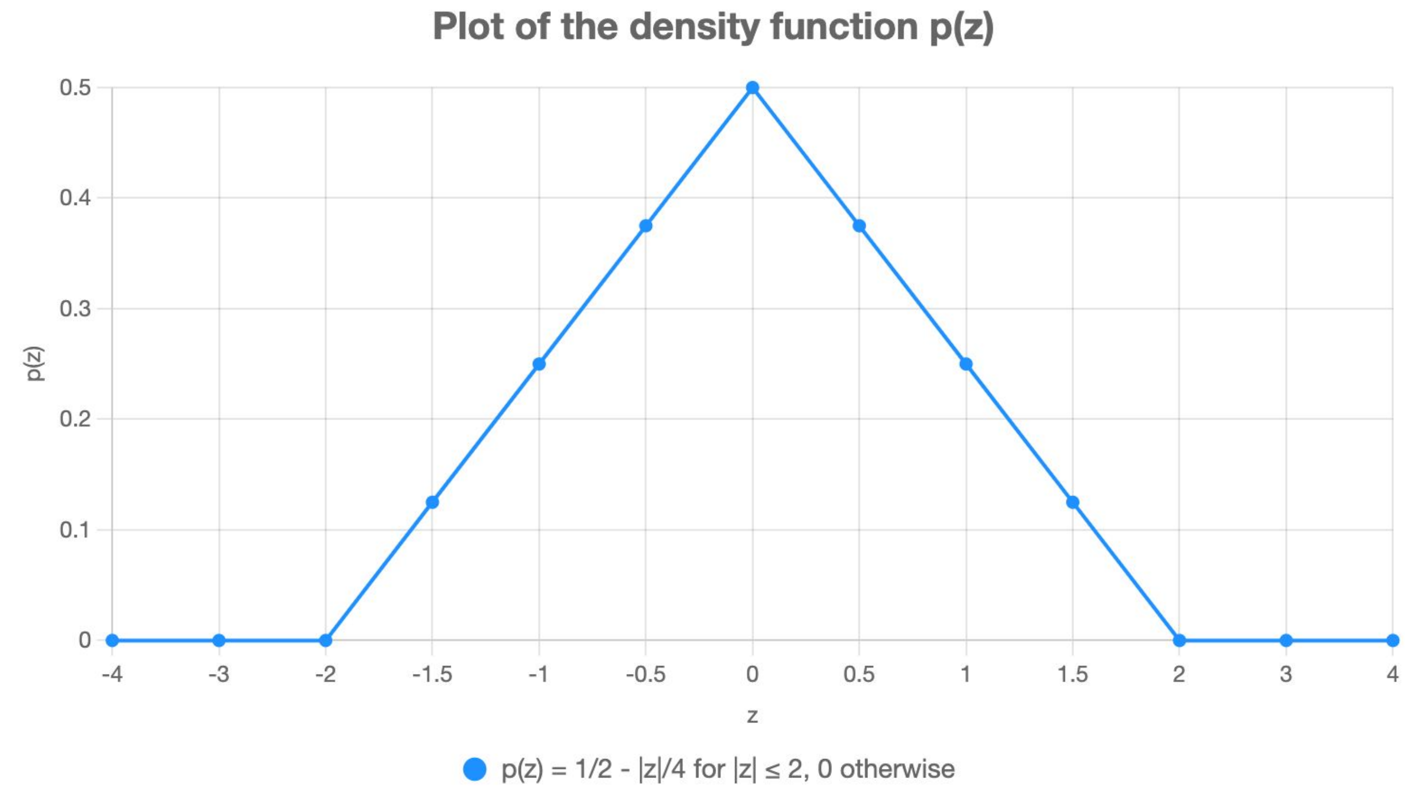
$$\varphi(t) \coloneqq \Pr[\boldsymbol{X} \leq t].$$

Suppose the density function is given by $p(z)$. Then,

it follows: $\varphi(t) = \int_{z=-\infty}^{t} p(z)\, dz.$

Example: Density function of r.v. $\boldsymbol{X}$ given by $p(z)$

$$p(z) = \begin{Bmatrix} \frac{1}{2} - \frac{|z|}{4} & \text{if} & |z| \leq 2 \\ 0 & \text{if} & |z| > 2 \end{Bmatrix}$$



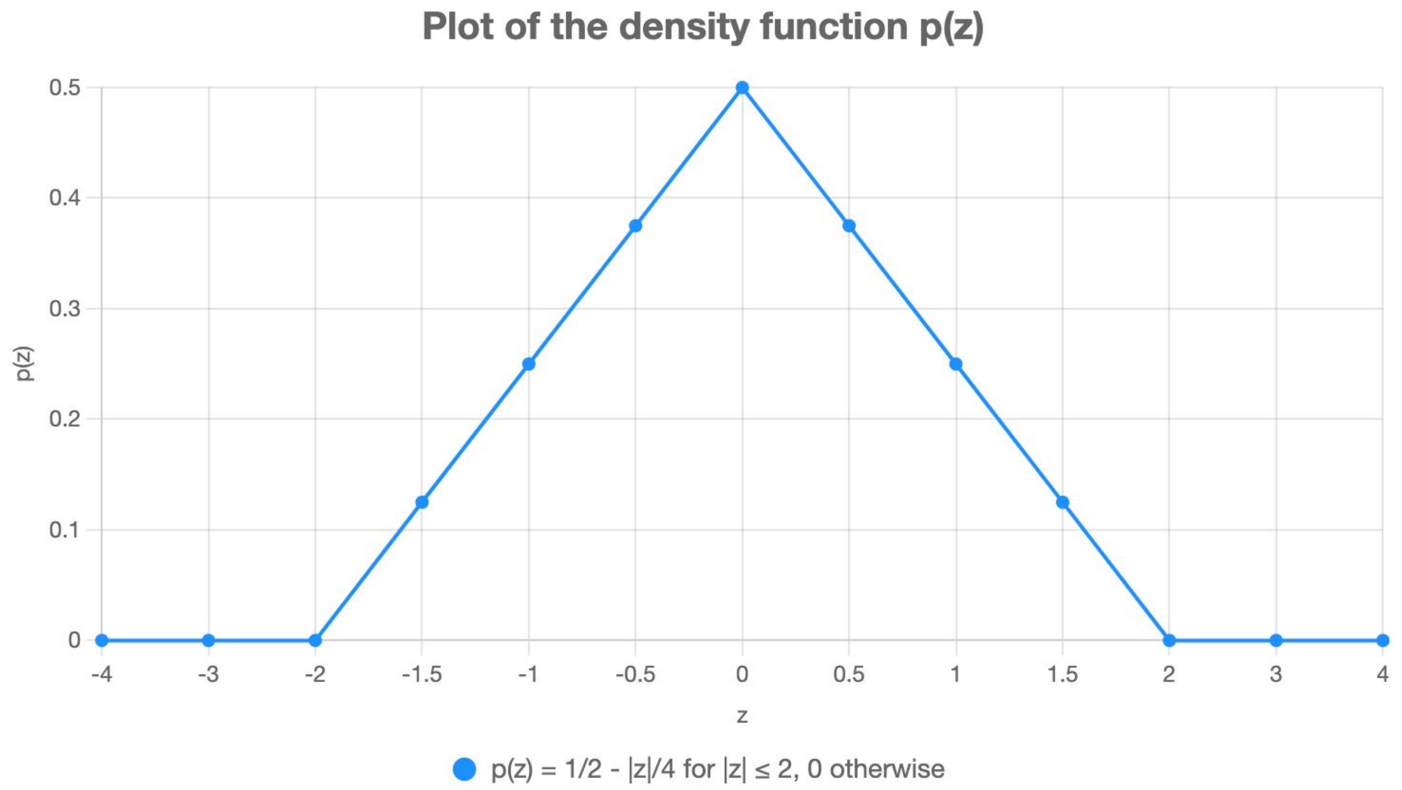
Cumulative distribution function

$$\varphi(t) = \Pr[\boldsymbol{X} \leq t] = \int_{z=-\infty}^{t} p(z)dz.$$

1. If $t \leq -2$, $\varphi(t) = 0$ (because $p(z) = 0$ if $z \leq -2$)

2. If -2 $\leq t \leq 0, \varphi(t) = \int_{z=-\infty}^{t} p(z)dz = \int_{z=-2}^{t} p(z)dz$

$$= \int_{z=-2}^{t} \left(\frac{1}{2} - \frac{|z|}{4}\right) dz = \int_{z=-2}^{t} \left(\frac{1}{2} + \frac{z}{4}\right) dz = \frac{1}{2} + \frac{t}{2} + \frac{t^2}{8}.$$



Cumulative distribution function

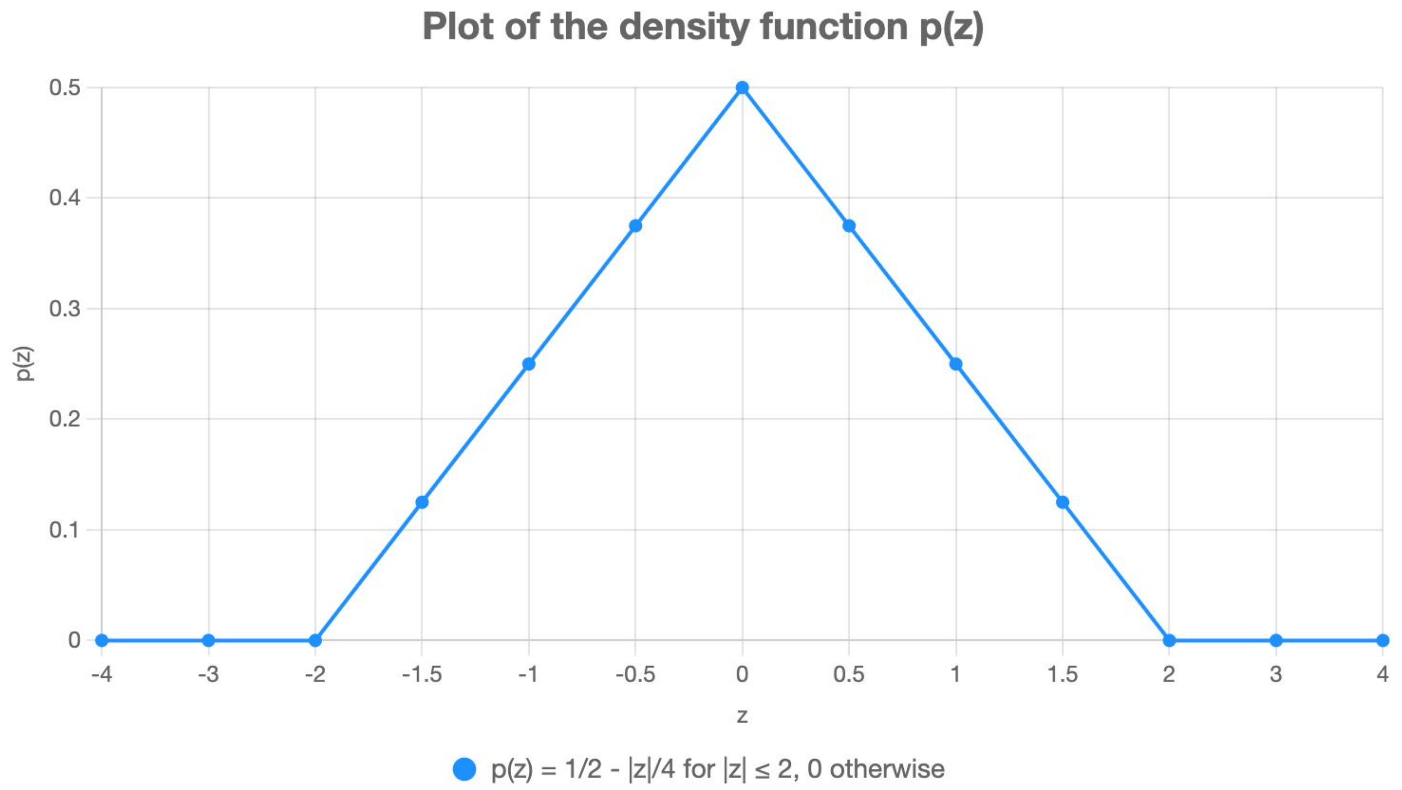
$$\varphi(t) = \Pr[\boldsymbol{X} \leq t] = \int_{z=-\infty}^{t} p(z)dz.$$

3. If $0 \leq t \leq 2, \varphi(t) = \int_{z=-\infty}^{t} p(z)dz$

$$= \int_{z=-\infty}^{0} p(z)dz + \int_{z=0}^{t} p(z)dz = \frac{1}{2} + \frac{t}{2} - \frac{t^2}{8}.$$

4. Finally, for $t \geq 2, \varphi(t) = \Pr[\boldsymbol{X} \leq t] = 1$

(pdf vanishes for any $z \geq 2$).



Relation between cdf and pdf

In summary, if the pdf of a random variable $\boldsymbol{X}$ is $p(\cdot)$ and the cdf is $\varphi(\cdot)$, then

$$\varphi(t) = \int_{z=-\infty}^{t} p(z)\, dz.$$

Thus, if $\varphi(\cdot)$ is differentiable, then $p(z) = \frac{d\varphi(z)}{dz}$, i.e., $p(\cdot)$ is the derivative of $\varphi(\cdot)$.

In many application, $\varphi(\cdot)$ is differentiable and thus, we can easily get the density function $p(\cdot)$ in terms of $\varphi(\cdot)$.

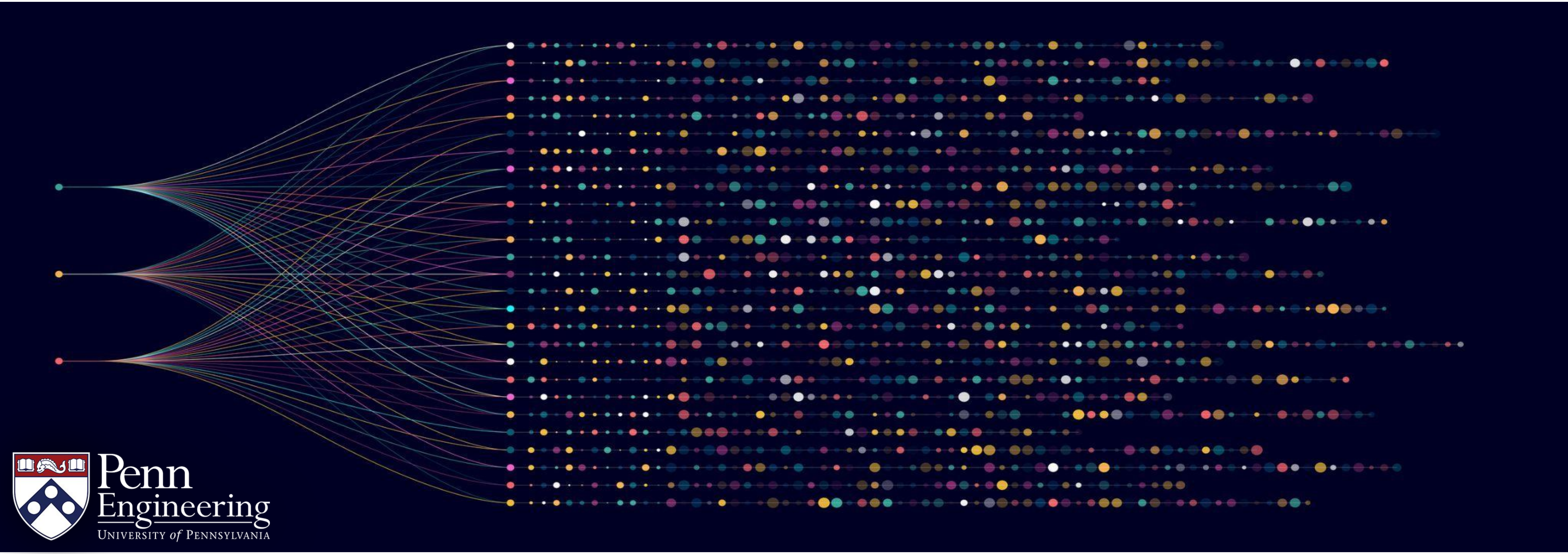
Continuous random variables: summary

- Continuous random variables occur at many places – theory and practice
- Specified by the probability density function pdf
- Dictionary between discrete and continuous worlds – summation replaced by integrals and point probabilities replaced by pdf

Algorithms for Big Data

Anindya De Erik Waingarten

The power of inclusion - exclusion



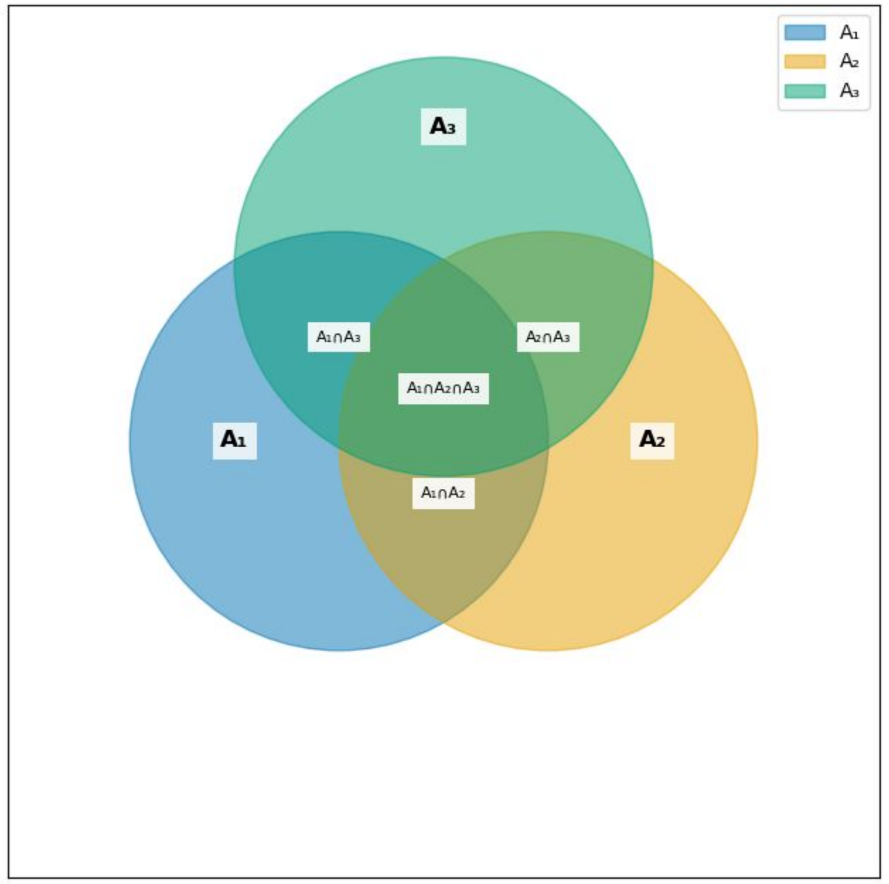
Inclusion exclusion principle

Elementary tool from combinatorics:

Inclusion-exclusion principle – used to reason about the size of the union of sets

Roughly speaking: the inclusion principle gives a series where we alternately overestimate and underestimate the size of the union

Inclusion Exclusion principle



Suppose we have three finite sets A_1, A_2, A_3

Inclusion exclusion principle says that

$|A_1 \cup A_2 \ \cup A_3| \leq \ |A_1| + |A_2| + |A_3| \ - \ \ (1).$

$|A_1 \cup A_2 \ \cup A_3| \geq \ |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3| \quad - (2).$

$|A_1 \cup A_2 \ \cup A_3| = \ |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3| + |A_1 \cap A_2 \cap A_3| \quad - (3).$

Equality in (3) because the number of sets here is 3. Inclusion exclusion can be extended to arbitrary number of sets

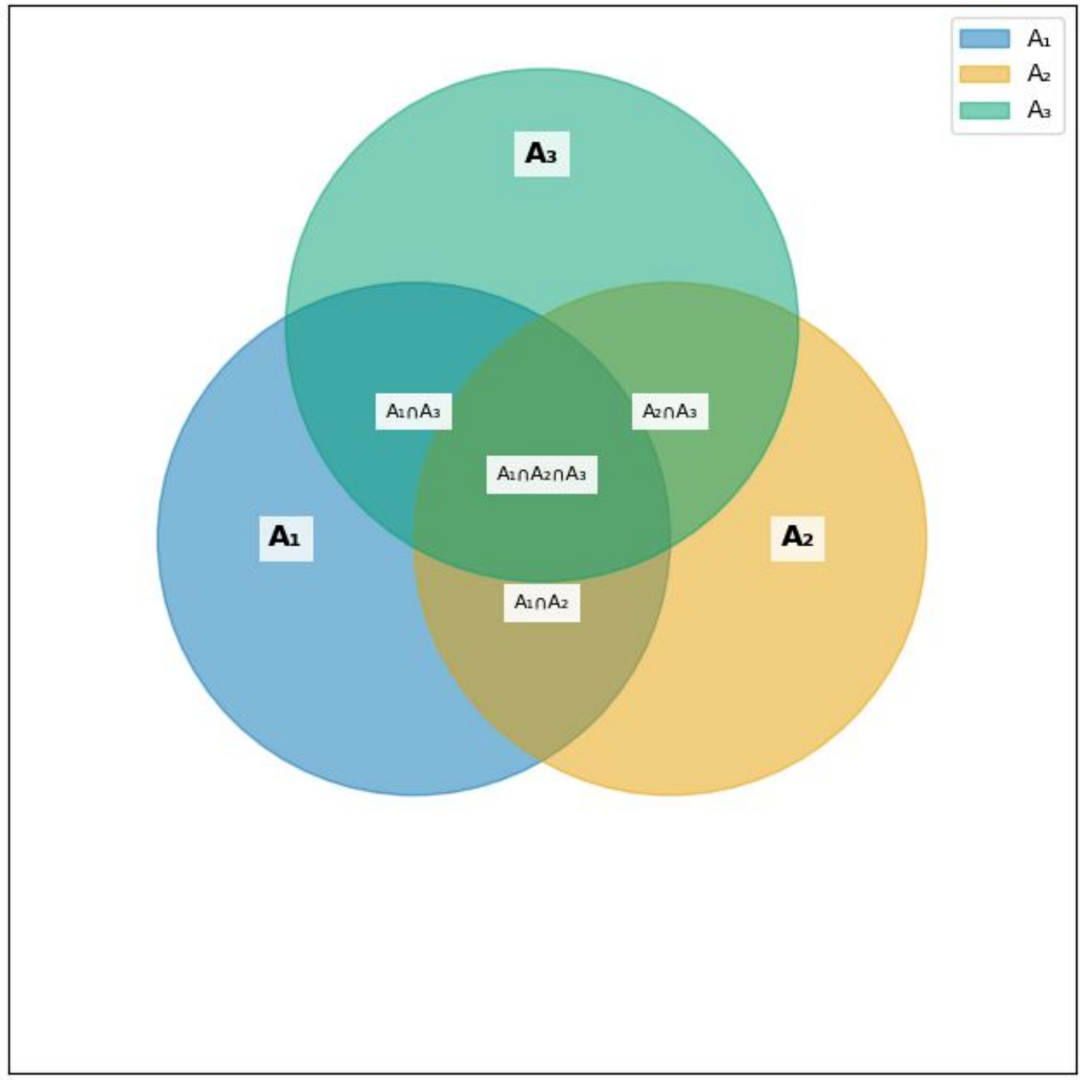
Inclusion Exclusion principle

Suppose we have three finite sets A_1, A_2, A_3.

Inclusion exclusion principle says that

$|A_1 \cup A_2 \cup A_3| \leq |A_1| + |A_2| + |A_3|$ – (1)

Size of the union is ≤ Sum of the sizes of the individual sets



$|A_1 \cup A_2 \cup A_3| \geq |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3|$ – (2).

Size of the union is ≥ Sum of the sizes of the individual sets – Sum of the sizes of pairwise intersections

Inclusion Exclusion principle

Suppose we have k finite sets $A_1, A_2, \dots, A_k$

Inclusion exclusion principle says that

$$|A_1 \cup A_2 \cup A_3 \dots \cup A_k| \leq \sum_{\ell=1}^{k} |A_\ell| \qquad -(1')$$

$$|A_1 \cup A_2 \cup A_3 \dots \cup A_k| \geq \sum_{\ell=1}^{k} |A_\ell| - \sum_{1 \leq j < \ell \leq k} |A_j \cap A_\ell| \qquad -(2').$$

Question: What does this have to do with probabilities?

Inclusion Exclusion principle for probability spaces

Suppose we have k events $E_1, \dots, E_k$.

Inclusion exclusion principle says that

$\Pr[E_1 \cup E_2 \cup \dots \cup E_k] \leq \sum_{\ell=1}^{k} \Pr[E_\ell]$ $\quad -(1'')$

$\Pr[E_1 \cup E_2 \cup \dots \cup E_k] \geq \sum_{\ell=1}^{k} \Pr[E_\ell] - \sum_{1 \leq j < \ell \leq k} \Pr[E_j \cap E_\ell]$ $\quad -(2'')$.

$E_1 \cup E_2 \cup \dots \cup E_k$: means at least one of the events $E_1, \dots, E_k$ occurs.

$E_j \cap E_\ell$: means both the events E_j and E_ℓ occur.

Inclusion Exclusion principle for probability spaces

Suppose we have k events $E_1, \dots, E_k$.

Inclusion exclusion principle says that

$$\Pr[E_1 \cup E_2 \cup \dots \cup E_k] \leq \sum_{\ell=1}^{k} \Pr[E_\ell] \qquad -(1'')$$

$$\Pr[E_1 \cup E_2 \cup \dots \cup E_k] \geq \sum_{\ell=1}^{k} \Pr[E_\ell] - \sum_{1 \leq j < \ell \leq k} \Pr\left[E_j \cap E_\ell\right] \qquad -(2'').$$

Equation ($1''$) is called the "Union bound" in probability theory – very powerful tool!

Application: Inclusion-Exclusion principle

Recall the application – computing fixed points of a random permutation

We have : $A = \{1,2,\ldots,n\}$ and let $\boldsymbol{B}$ be a random permutation of A.

We are interested in the number of fixed points of $\boldsymbol{B}$ (denoted by $\mathrm{FP}(\boldsymbol{B})$)

– i.e., positions i which are unchanged in $\boldsymbol{B}$.

Previously: we have shown $\mathrm{E}[\mathrm{FP}(\boldsymbol{B})] = 1$ and $\Pr[\mathrm{FP}(\boldsymbol{B}) \geq c] \leq \frac{1}{c}$.

Question: What is the probability that $\mathrm{FP}(\boldsymbol{B}) \geq 1$?

Number of fixed points of a random permutation

We have shown $\mathrm{E}[\mathrm{FP}(\boldsymbol{B})] = 1$ and $\Pr[\mathrm{FP}(\boldsymbol{B}) \geq c] \leq \frac{1}{c}$.

What is the probability that $\mathrm{FP}(\boldsymbol{B}) \geq 1$? Is it at least *say* 0.1?

Scenario 1: $\Pr[\mathrm{FP}(\boldsymbol{B}) = 0] = 1 - \frac{1}{\mathrm{n}}$ and $\Pr[\mathrm{FP}(\boldsymbol{B}) = n] = \frac{1}{n}$?

Note that in scenario #1, $\Pr[\mathrm{FP}(\boldsymbol{B}) \geq 1] = \frac{1}{n}$.

We will show that Scenario 1 is not possible!

Lower bounding the probability of having at least one fixed point

Recall our notation from previous video:

$FP_j(\boldsymbol{B}) = 1$ if j is a fixed point in $\boldsymbol{B}$ and 0 otherwise.

Define the event E_j to be when $FP_j(\boldsymbol{B}) = 1$.

Observation #1: The event $\mathrm{FP}(\boldsymbol{B}) \geq 1$ is the same as the event $E_1 \cup E_2 \ \ldots \cup E_n$.

Applying Inclusion Exclusion

$FP_j(\boldsymbol{B}) = 1$ if j is a fixed point in $\boldsymbol{B}$ and 0 otherwise.

Define the event E_j to be when $FP_j(\boldsymbol{B}) = 1$.

So, by equation (2") of Inclusion-Exclusion, we have

$$\Pr[E_1 \cup E_2 \cup \ldots \cup E_n] \geq \sum_{\ell=1}^{n} \Pr[E_\ell] - \sum_{1 \leq j < \ell \leq n} \Pr\left[E_j \cap E_\ell\right] \quad -(4).$$

Note that $\Pr[E_\ell] = \Pr[FP_\ell(\boldsymbol{B}) = 1] = \frac{1}{n}$ $-(5)$

Calculating probabilities of pairwise intersection

What about the probabilities of pairwise intersections?

$\Pr[E_j \cap E_\ell]$ = probability that both j and ℓ are fixed points

A random permutation is equally likely to map j and ℓ to any of $n(n-1)$ possibilities. Thus, the probability that j is mapped to j and ℓ is mapped to ℓ is exactly $\frac{1}{n(n-1)}$.

Thus, $\Pr[E_j \cap E_\ell] = \frac{1}{n(n-1)} \qquad - (6).$

Putting it all together

Recall equation (4) says

$$\Pr[E_1 \cup E_2 \ \cup \ \ldots \cup E_n] \geq \sum_{\ell=1}^{n} \Pr[E_\ell] - \sum_{1 \leq j < \ell \leq n} \Pr\left[E_j \cap E_\ell\right].$$

From equation (5), $\Pr[E_\ell] = \frac{1}{n}$ and equation (6), $\Pr\left[E_j \cap E_\ell\right] = \frac{1}{n(n-1)}$.

$$\Pr[E_1 \cup E_2 \ \cup \ \ldots \cup E_n] \geq \sum_{\ell=1}^{n} \frac{1}{n} - \sum_{1 \leq j < \ell \leq n} \frac{1}{n(n-1)} = n \cdot \frac{1}{n} - \binom{n}{2} \cdot \frac{1}{n(n-1)} = 1 - \frac{1}{2} = \frac{1}{2}.$$

Summary

Thus, we can conclude that for a random permutation $\Pr[\mathrm{FP}(\boldsymbol{B}) \geq 1] \geq \frac{1}{2}$.

More generally, inclusion exclusion principle can be a powerful tool to analyze probabilistic experiments– especially establishing lower bounds on probabilities